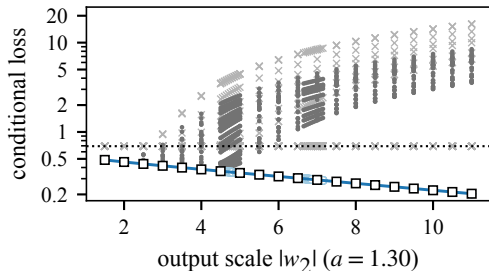
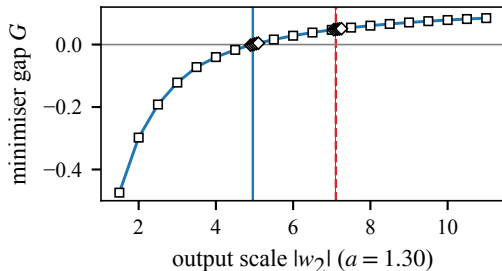


- × degenerate (discarded): 1,194 — retained branch (39 scales) ◇ stricter optimisation, 1e (10)
- screening runs: 976 constant predictor (log 2) — certified R_{glob} bracket
- full runs, not lowest: 413 □ certified global min. (20) - - - certified R_{solve} bracket

(a) every candidate of the frozen search



(b) the branch crosses $G = 0$ continuously



(a) $a = 1.30$: $n = 2,701$ candidates at 59 evaluations; frozen result reproduced 59/59; none below the retained branch (full runs not lowest lie on it); certified widths $\leq 1\text{e-}09$.

(b) certified brackets: R_{glob} at $|w_2| \in (4.9500, 4.9625]$, R_{solve} at $(7.1000, 7.1125]$; G changes sign once; minimisers either side of the switch $\leq 2.7\text{e-}04$ apart (certified); other basins $\geq +0.0031$ above (certified, radius 0.1).